2.2 Theorems about Parallel Lines

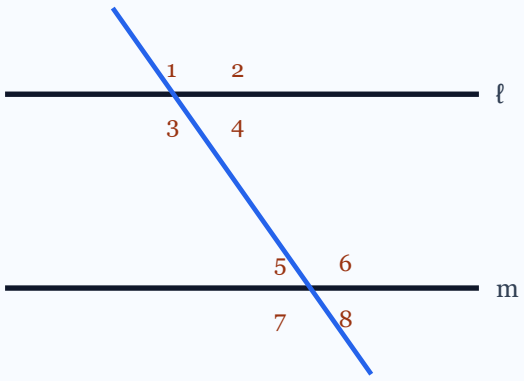
Once two lines are known to be parallel, every angle a transversal creates is locked into place: corresponding, alternate interior, and alternate exterior pairs are congruent, while same-side interior pairs are supplementary — never the other way around, which is a common mix-up. This section also runs the logic backward with the *converse* theorems, which let you prove lines are parallel from an angle relationship instead of assuming it. Expect multi-step algebra, a brief look at why the theorems follow from the Corresponding Angles Postulate plus Vertical Angles, and problems that chain several angle facts together.

Examples

Example 1: Corresponding Angles Postulate

Lines $\ell \parallel m$ are cut by a transversal. Corresponding angles measure $3x$ and $75°$. Find x.

Solution



ℓ ∥ m — corresponding angles are congruent.

Postulate: when $\ell \parallel m$, corresponding angles are congruent (Corresponding Angles Postulate).

Set the two measures equal: $3x = 75$.

Answer: $x = 25$ (both angles measure $75°$).

Example 2: Alternate Interior Angles Theorem

$\ell \parallel m$, cut by a transversal. Alternate interior angles measure $2x + 10$ and $70°$. Find x, and briefly explain why the theorem is true.

Solution

Theorem: alternate interior angles are congruent when $\ell \parallel m$. This follows from the Corresponding Angles Postulate plus the Vertical Angles Theorem — the alternate-interior angle is vertical to the angle corresponding to the other one, so by the Transitive Property all three are congruent.

Set the measures equal: $2x + 10 = 70$.

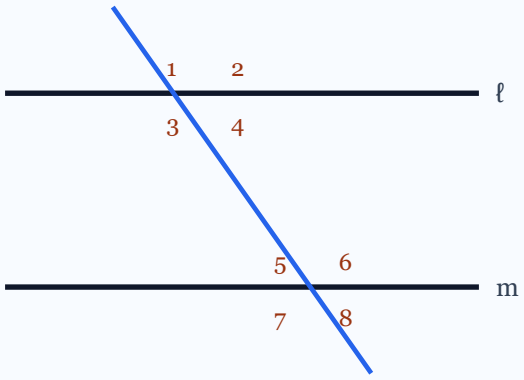
$2x = 60 \Rightarrow x = 30$.

Answer: $x = 30$ (both angles measure $70°$).

Example 3: Same-Side Interior Angles Theorem

$\ell \parallel m$, cut by a transversal. Same-side interior angles measure $4x$ and $2x + 30$. Find x.

Solution



Same-side interior angles are supplementary, not congruent, when ℓ ∥ m.

Theorem: same-side (consecutive) interior angles are *supplementary* when $\ell \parallel m$ — this is the one relationship in the chapter that is **not** congruence.

$4x + (2x + 30) = 180$

$6x + 30 = 180 \Rightarrow 6x = 150 \Rightarrow x = 25$.

Check: $4(25) = 100°$ and $2(25) + 30 = 80°$; $100° + 80° = 180°$. ✓

Example 4: Alternate Exterior Angles Theorem

$\ell \parallel m$, cut by a transversal. Alternate exterior angles both equal $x + 20$ and $3x - 40$. Find x.

Solution

Theorem: alternate exterior angles are congruent when $\ell \parallel m$.

Set the measures equal: $x + 20 = 3x - 40$.

$60 = 2x \Rightarrow x = 30$.

Check: both angles equal $30 + 20 = 50°$. ✓

Example 5: Direct application

$\ell \parallel m$; one angle is $112°$. Find its corresponding angle.

Solution

By the Corresponding Angles Postulate, corresponding angles are congruent whenever the lines are parallel — no algebra needed once the relationship is identified.

Answer: $112°$.

Example 6: Same-side supplement

$\ell \parallel m$; one same-side interior angle is $65°$. Find the other.

Solution

By the Same-Side Interior Angles Theorem, the two measures must sum to $180°$.

$65° + x = 180° \Rightarrow x = 115°$.

Answer: $115°$.

Example 7: Using the converse to prove parallel

A transversal crosses lines j and k, creating a pair of corresponding angles that both measure $58°$. What can you conclude about j and k, and which theorem justifies it?

Solution

This problem runs the logic backward from Examples 1–6: instead of *starting* from parallel lines to find equal angles, we *start* from equal angles to prove the lines are parallel.

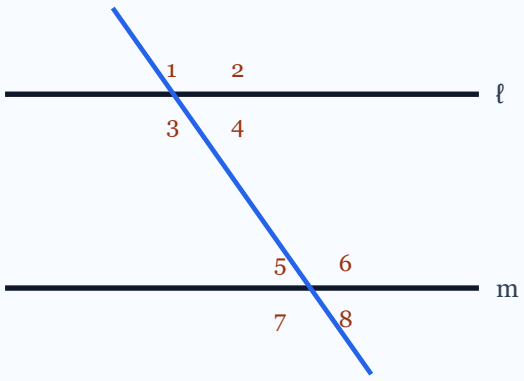
Converse of the Corresponding Angles Postulate: if corresponding angles formed by a transversal are congruent, the two lines are parallel.

Answer: $j \parallel k$ (by the Converse of the Corresponding Angles Postulate).

Example 8: Chaining vertical, corresponding, and linear pair

$\ell \parallel m$, cut by transversal t. One angle at the ℓ intersection measures $50°$. Find (a) its vertical angle, (b) its corresponding angle at the m intersection, and (c) the linear-pair partner of that corresponding angle.

Solution



One value at ℓ propagates to every other angle in the figure.

(a) Vertical Angles Theorem: the vertical angle at the same intersection is congruent, so it is $50°$.

(b) Corresponding Angles Postulate ($\ell \parallel m$): the corresponding angle at m is also $50°$.

(c) Linear Pair Postulate: the linear-pair partner is supplementary to that $50°$ angle: $180° - 50° = 130°$.

Answer: (a) $50°$ (b) $50°$ (c) $130°$.

Example 9: Linear pair then corresponding

$\ell \parallel m$; one angle at the ℓ intersection is $130°$. Find its linear-pair partner, then find that partner's corresponding angle at m.

Solution

Linear Pair Postulate: the partner angle is supplementary: $180° - 130° = 50°$.

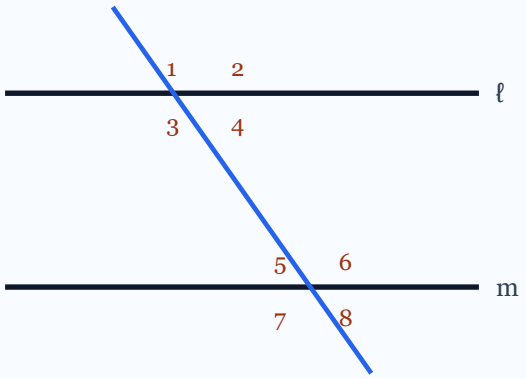
Corresponding Angles Postulate: the corresponding angle to that $50°$ angle at line m is also $50°$.

Answer: Linear-pair partner $= 50°$; its corresponding angle at $m = 50°$.

Example 10: Solve for x using the converse

A transversal creates an angle of $(2x + 5)°$ with line ℓ and the corresponding angle of $(3x - 15)°$ with line m. For what value of x are ℓ and m guaranteed parallel, and what is the common angle measure?

Solution



Corresponding angles equal exactly when the lines are parallel.

Converse of the Corresponding Angles Postulate: the lines are parallel exactly when the corresponding angles are equal.

Set the expressions equal: $2x + 5 = 3x - 15$.

$20 = x$.

Answer: $x = 20$; each angle measures $2(20) + 5 = 45°$.

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Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Corresponding

Parallel; corresponding $5x$ and $80°$. Find x.

Answer: **Verification:**

Problem 2: Alternate interior

Parallel; alt. interior $x + 15$ and $55°$. Find x.

Answer: **Verification:**

Problem 3: Same-side interior

Parallel; same-side $3x$ and $x + 40$. Find x.

Answer: **Verification:**

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Problem 4: Alternate exterior

Parallel; alt. exterior $2x$ and $x + 28$. Find x.

Answer: **Verification:**

Problem 5: Congruent measure

Parallel; corresponding to $95°$?

Answer: **Verification:**

Problem 6: Supplementary

Parallel; same-side to $70°$?

Answer: **Verification:**

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Problem 7: Converse

Alt. interior angles are congruent. What can you conclude about the lines?

Answer: **Verification:**

Problem 8: Find x

Parallel; corresponding $4x - 8$ and $2x + 20$. Find x.

Answer: **Verification:**

Problem 9: Same-side

Parallel; same-side $5x + 10$ and $4x$. Find x.

Answer: **Verification:**

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Problem 10: Vertical

Angle $48°$; its vertical is?

Answer: **Verification:**

Problem 11: Linear pair

Angle $135°$; linear-pair partner?

Answer: **Verification:**

Problem 12: Alternate interior

Parallel; alt. interior $7x - 5$ and $5x + 17$. Find x.

Answer: **Verification:**

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Problem 13: Corresponding algebra

Parallel; corresponding $3x + 6$ and $4x - 12$. Find x.

Answer: **Verification:**

Problem 14: Challenge — converse

A transversal makes an angle of $(5x - 10)°$ with ℓ and the corresponding angle $(3x + 24)°$ with m. Find x, and give the common angle measure that guarantees $\ell \parallel m$.

Answer: **Verification:**

Problem 15: Challenge — same-side plus linear pair

$\ell \parallel m$. A same-side interior pair measures $4x$ and $2x + 12$. Find x, both angle measures, and the linear-pair partner of the larger angle.

Answer: **Verification:**

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✓ Answer Key

1. $x = 16$
2. $x = 40$
3. $x = 35$
4. $x = 28$
5. $95°$
6. $110°$
7. Parallel (Converse of the Alternate Interior Angles Theorem)
8. $x = 14$
9. $x = 20$
10. $48°$
11. $45°$
12. $x = 11$
13. $x = 18$
14. $x = 17$; common angle $= 75°$
15. $x = 28$; angles $112°$ and $68°$; linear-pair partner of $112°$ is $68°$

Great work — keep practising!